

WEEK – 01

Leet Code Problems on GCD

1. Find Greatest Common Divisor of Array.

Example 1:

Input: nums = [2,5,6,9,10]

Output: 2

Explanation:

The smallest number in nums is 2.

The largest number in nums is 10.

The greatest common divisor of 2 and 10 is 2.

```
</> Code
C v Auto

1  int gcd(int a, int b) {
2      while (b != 0) {
3          int temp = b;
4          b = a % b;
5          a = temp;
6      }
7      return a;
8  }
9
10 int findGCD(int* nums, int numsSize) {
11     int min = nums[0];
12     int max = nums[0];
13
14
15     for (int i = 1; i < numsSize; i++) {
16         if (nums[i] < min) {
17             min = nums[i];
18         }
19         if (nums[i] > max) {
20             max = nums[i];
21         }
22     }
23
24
25     return gcd(min, max);
26 }
```

Output:

☒ Testcase | ☒ Test Result

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ Case 3

Input

nums =
[2,5,6,9,10]

Output

2

Expected

2

2.Greatest Common Divisor of Strings.

Example 1:

Input: str1 = "ABCABC", str2 = "ABC"

Output: "ABC"

```
Code
C v Auto
1 #include <string.h>
2 #include <stdlib.h>
3
4 int gcd(int a, int b) {
5     return b == 0 ? a : gcd(b, a % b);
6 }
7
8 char* gcdOfStrings(char* str1, char* str2) {
9     int len1 = strlen(str1);
10    int len2 = strlen(str2);
11
12
13    char *temp1 = (char*)malloc(len1 + len2 + 1);
14    char *temp2 = (char*)malloc(len1 + len2 + 1);
15
16    strcpy(temp1, str1);
17    strcat(temp1, str2);
18
19    strcpy(temp2, str2);
20    strcat(temp2, str1);
21
22    if (strcmp(temp1, temp2) != 0) {
23        free(temp1);
24        free(temp2);
25        char *res = (char*)malloc(1);
26        res[0] = '\0';
27        return res;
28    }
29
30    free(temp1);
31    free(temp2);
32
33
34    int g = gcd(len1, len2);
35
36
37    char *result = (char*)malloc(g + 1);
38    strncpy(result, str1, g);
39    result[g] = '\0';
40
41    return result;
42 }
```

Output

☒ Testcase | [Test Result](#)

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ Case 3 ☒ Case 4

Input

```
str1 =  
"ABCABC"
```

```
str2 =  
"ABC"
```

Output

```
"ABC"
```

Expected

```
"ABC"
```

2. GCD of Odd and Even Sums.

Example 1:

Input: $n = 4$

Output: 4

Explanation:

- Sum of the first 4 odd numbers $\text{sumOdd} = 1 + 3 + 5 + 7 = 16$
- Sum of the first 4 even numbers $\text{sumEven} = 2 + 4 + 6 + 8 = 20$

Hence, $\text{GCD}(\text{sumOdd}, \text{sumEven}) = \text{GCD}(16, 20) = 4$.

[Code](#)

C Auto

```
1  int gcd(int a, int b) {  
2      while (b != 0) {  
3          int temp = b;  
4          b = a % b;  
5          a = temp;  
6      }  
7      return a;  
8  }  
9  int gcdOfOddEvenSums(int n) {  
10     int sumOdd = n * n;  
11     int sumEven = n * (n + 1);  
12     return gcd(sumOdd, sumEven);  
13 }
```

Output:

☒ Testcase |  Test Result

Accepted Runtime: 0 ms

☒ Case 1

☒ Case 2

Input

n =
4

Output

4

Expected

4